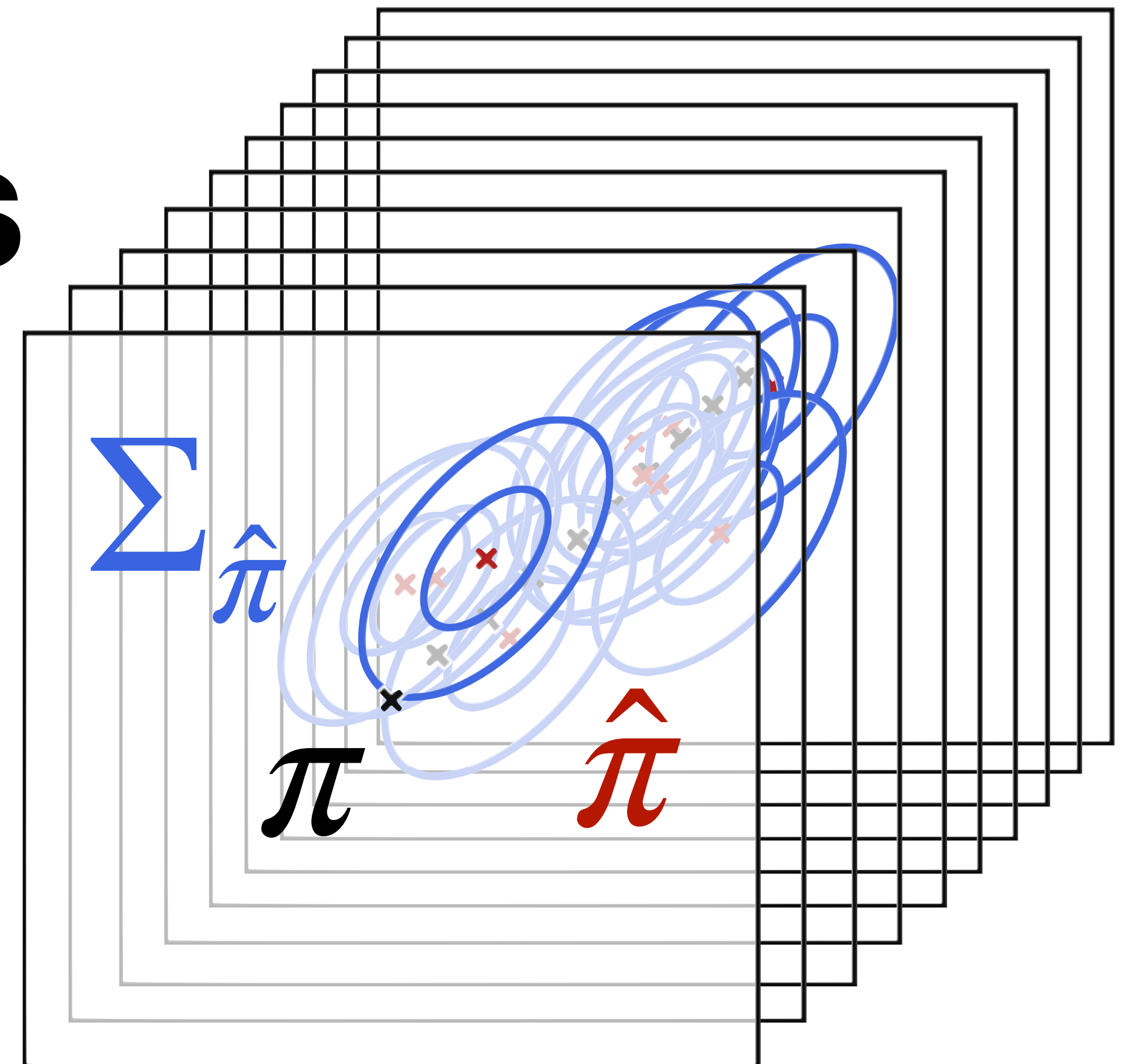


Frequentist Analysis

Jed Homer LMU | ODSL



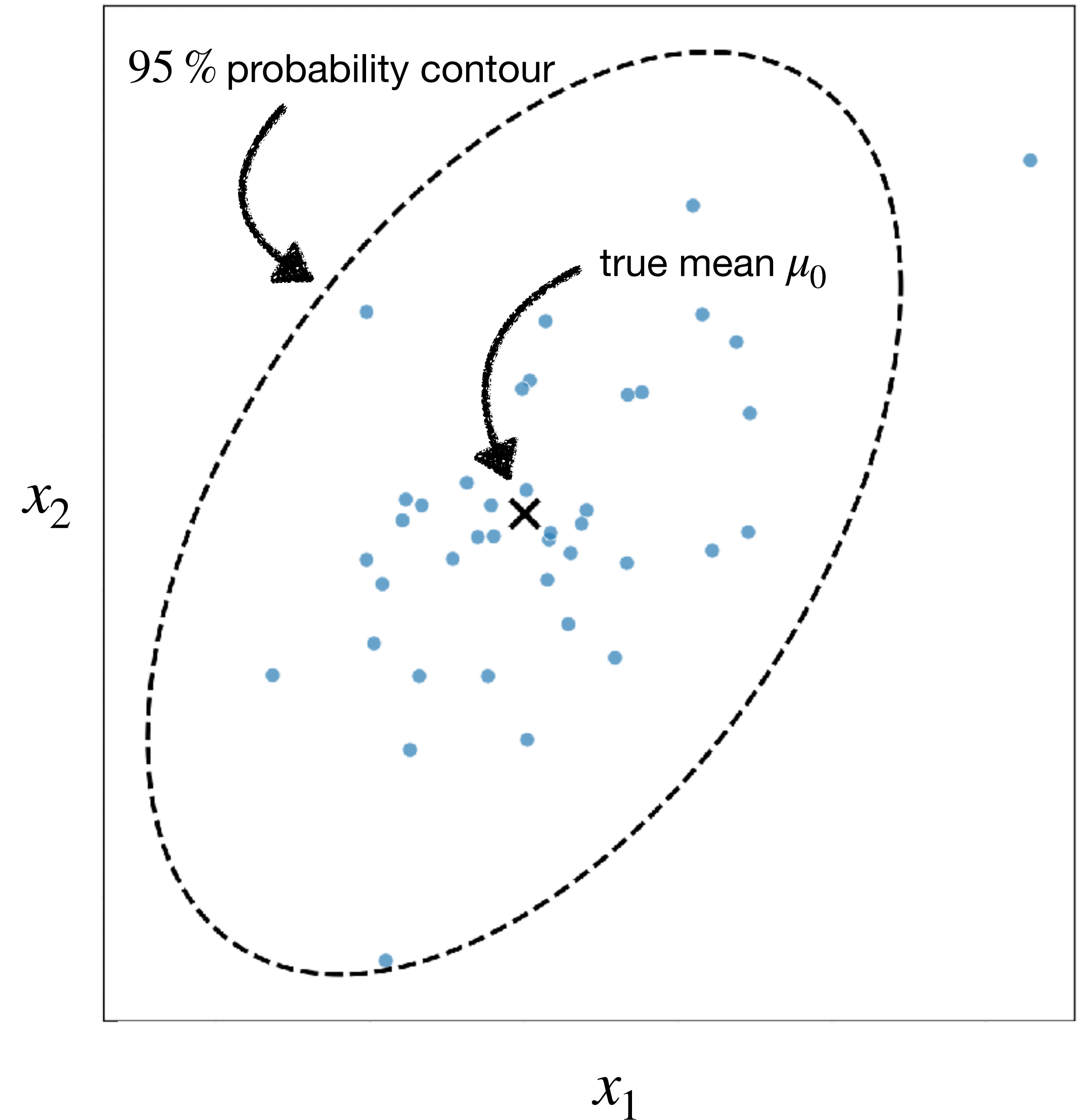
Today

- How does a Frequentist interpret probability?
- Let's look at what Bayesian and Frequentist analyses look like in practice.
- **No tutorial today, we'll look at Bayes / Freq. analyses in tutorial tomorrow!**

Setup

- We observe $\mathcal{X} = \{x_1, \dots, x_n\}$ where $x_i \sim \mathcal{G}[x | \mu, \Sigma]$.
- We know Σ .
- We want to infer μ from $\mathcal{X}$.

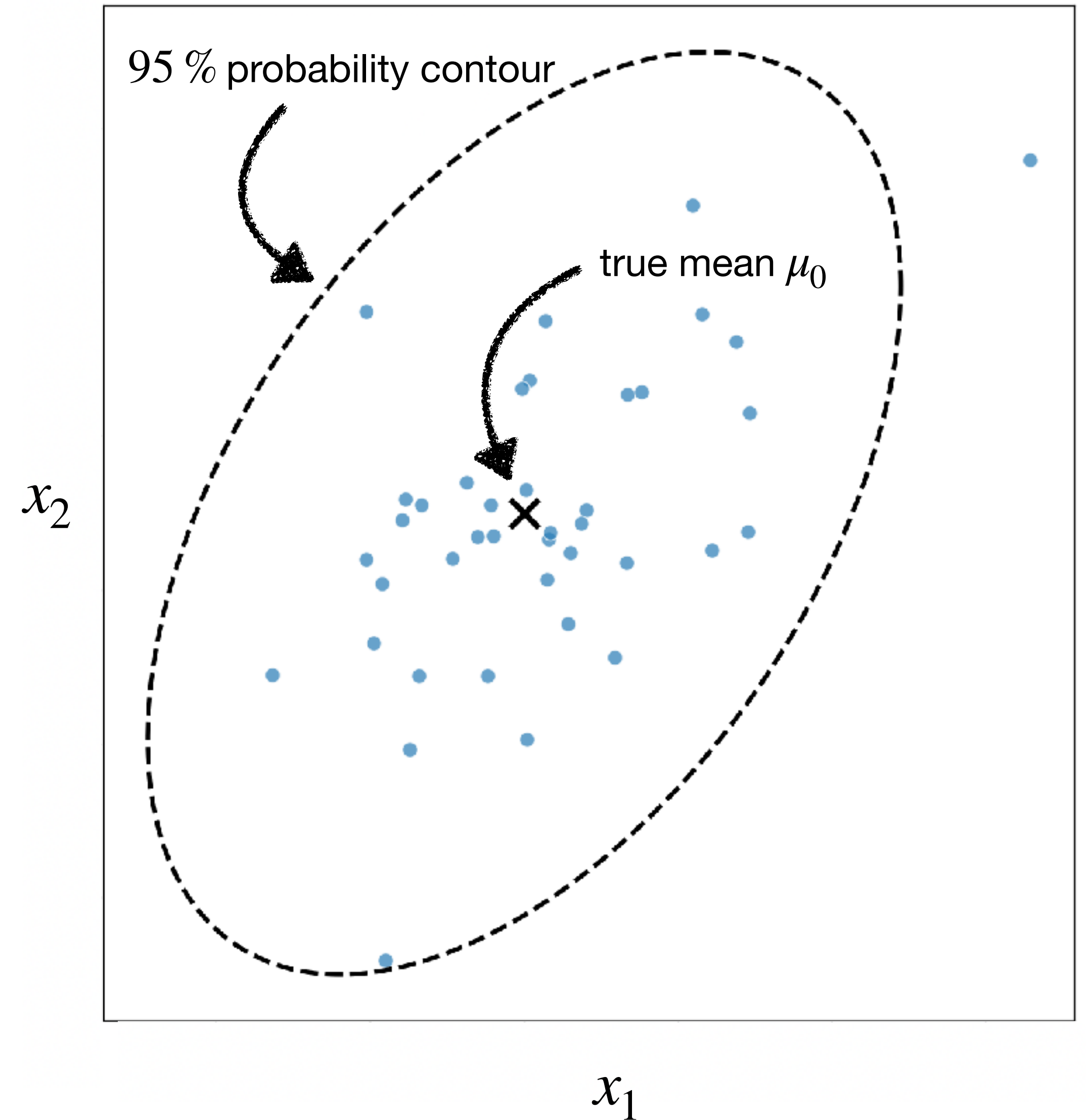
Samples $\mathcal{X} = \{x_1, \dots, x_n\} \sim \mathcal{G}(x | \mu, \Sigma)$



Quick detour: χ^2

- What does the contour in this plot mean?

Samples $\mathcal{X} = \{x_1, \dots, x_n\} \sim \mathcal{G}(x | \mu, \Sigma)$



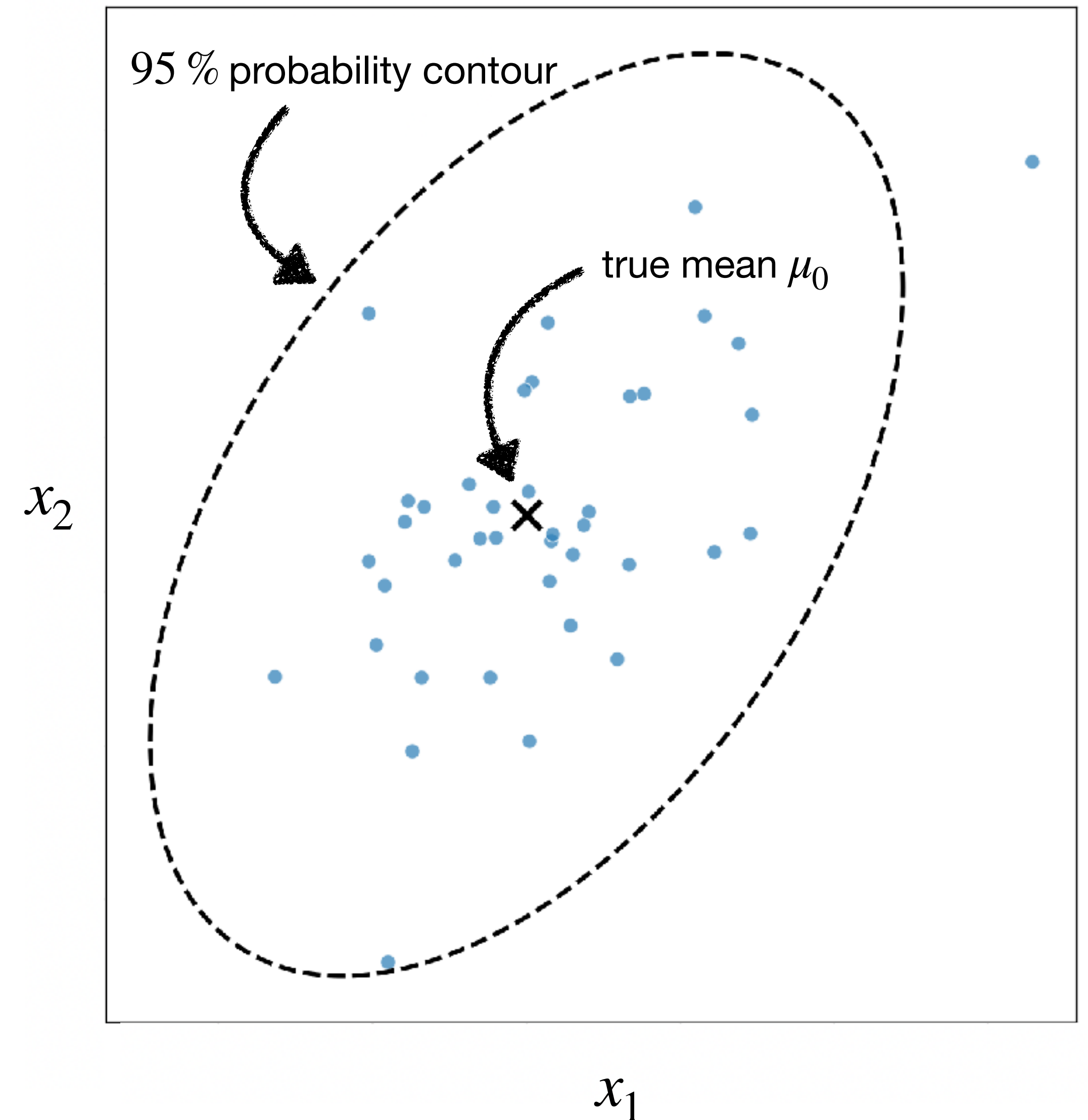
Quick detour: χ^2

- What does the contour in this plot mean?
- The scatter of a sampled point x from the true mean μ_0 is

$$(x - \mu_0)^T \Sigma^{-1} (x - \mu_0)$$

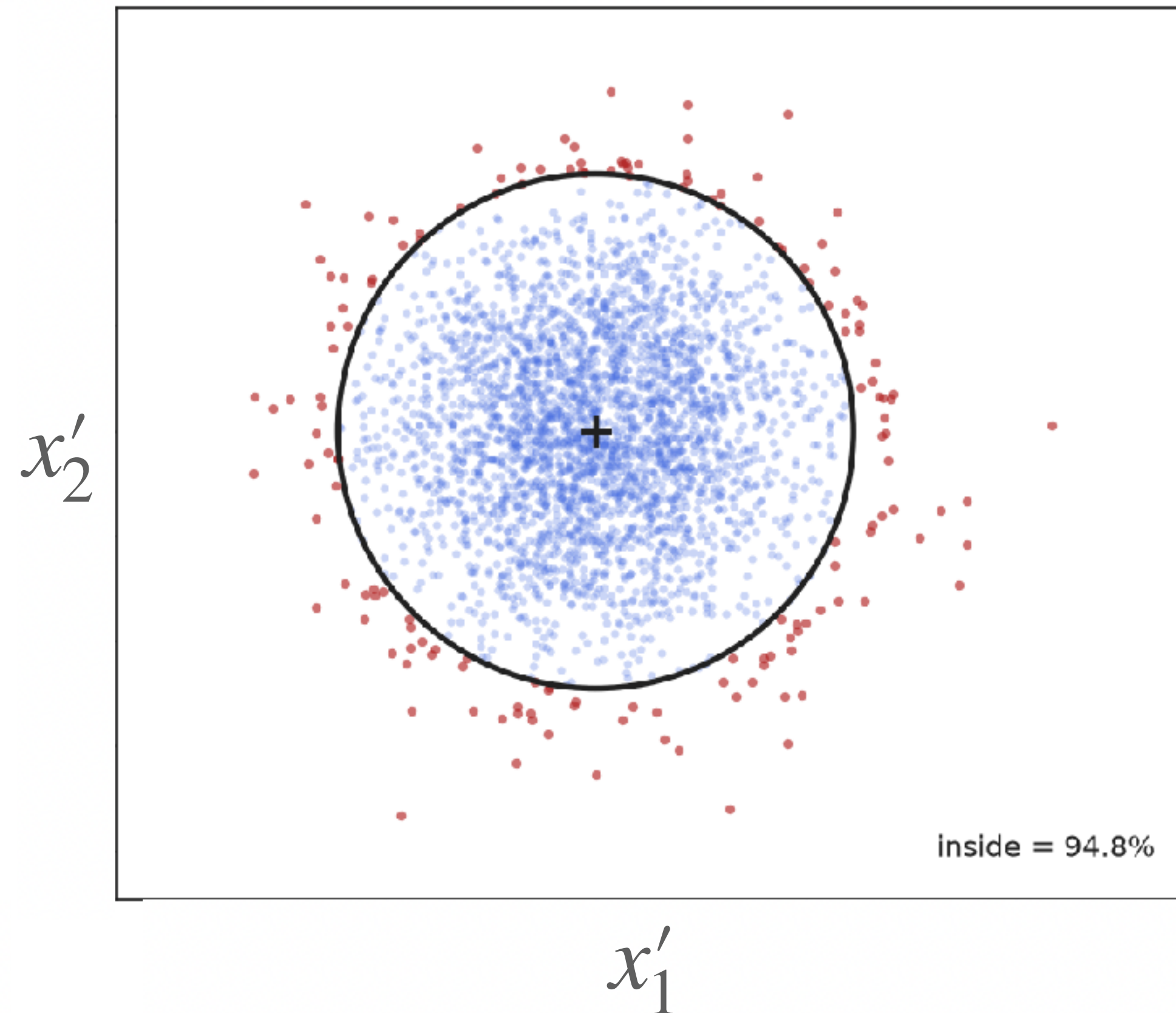
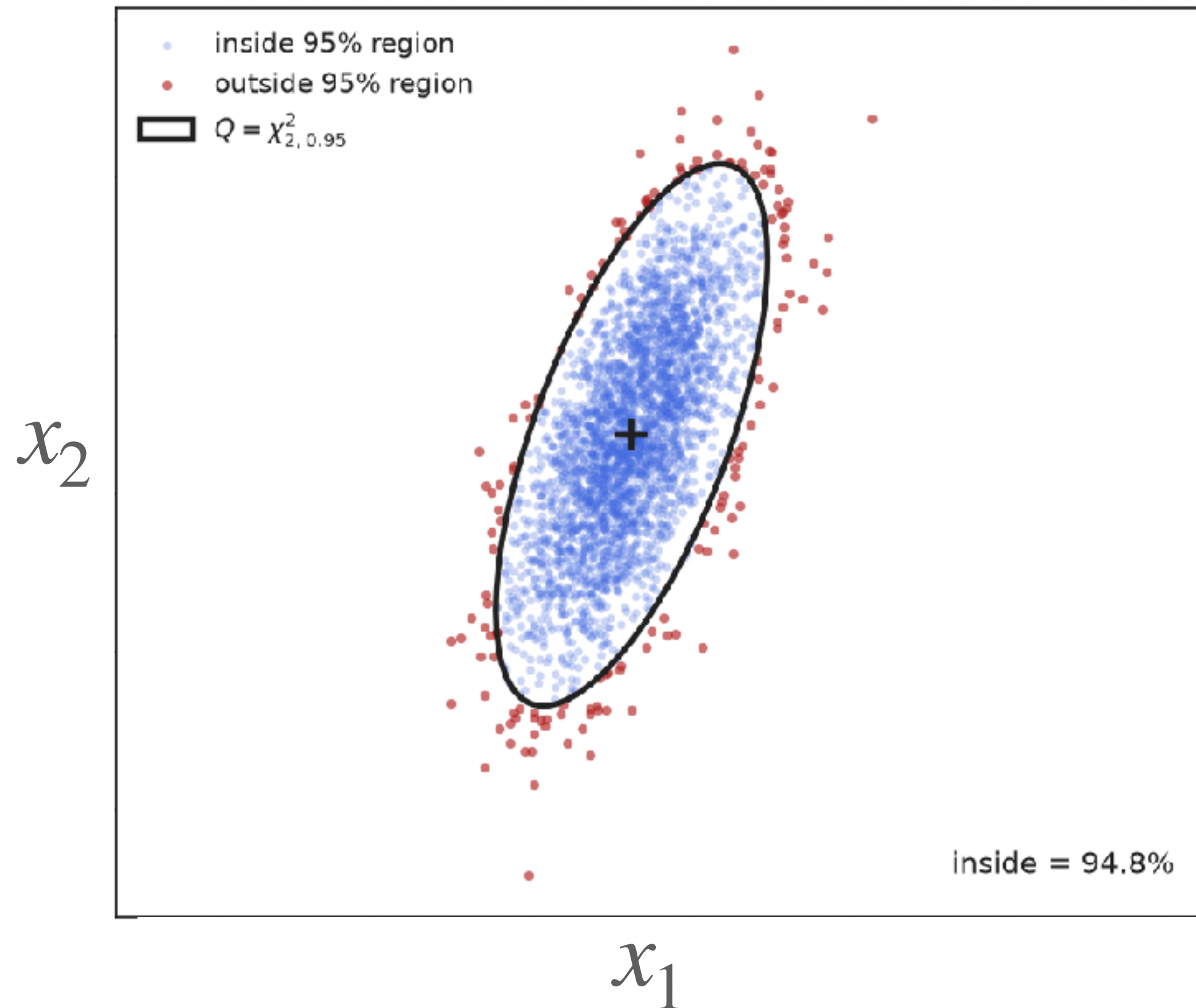
- What PDF does this quantity follow?

Samples $\mathcal{X} = \{x_1, \dots, x_n\} \sim \mathcal{G}(x | \mu, \Sigma)$



What is $(x - \mu)^T \Sigma^{-1} (x - \mu)$?

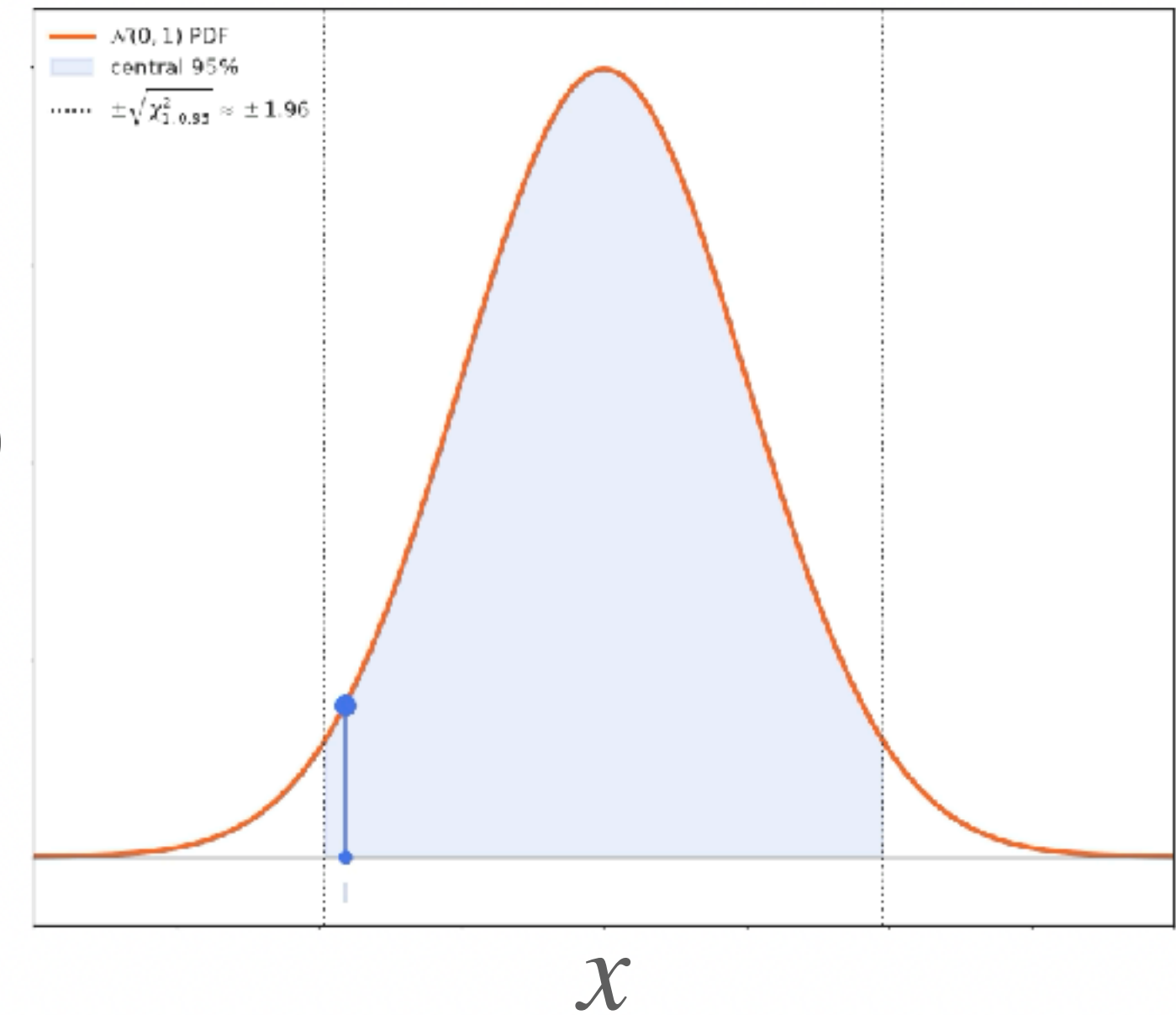
- This quadratic form just accounts for the location and correlations of x !



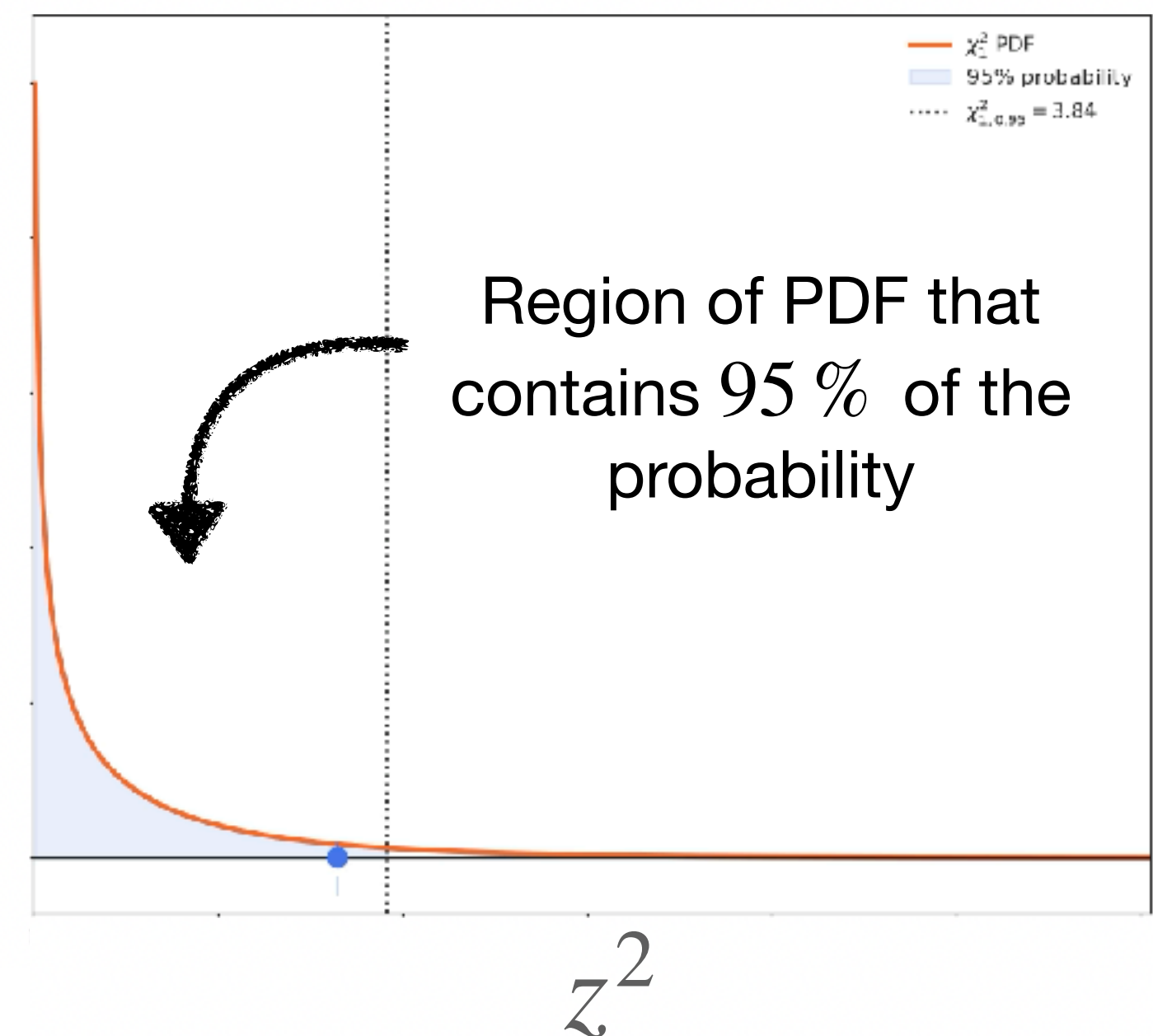
Quick detour: χ^2

- Let $x \sim \mathcal{G}[x | \mu, \sigma^2]$,

$p(x)$

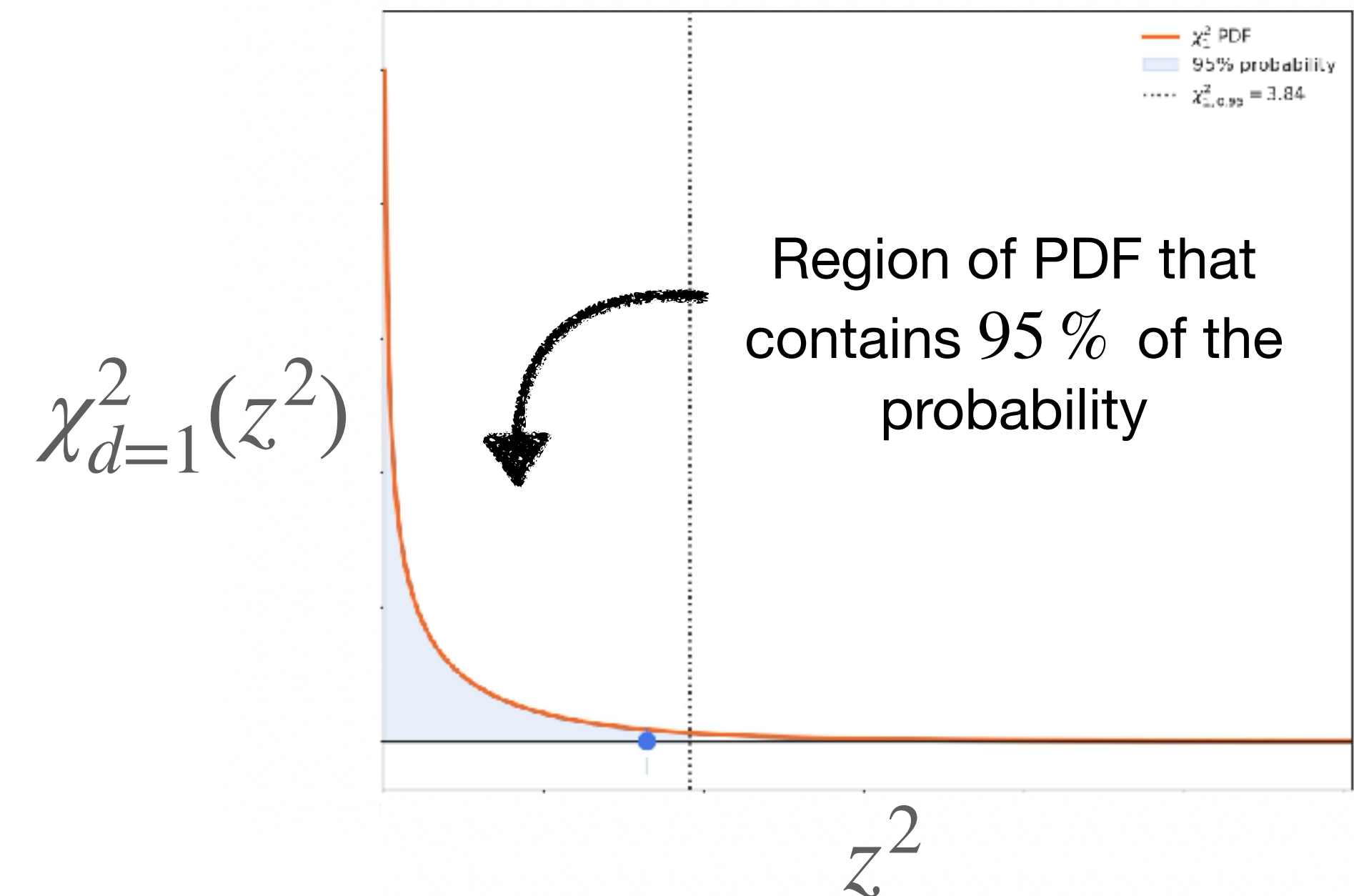
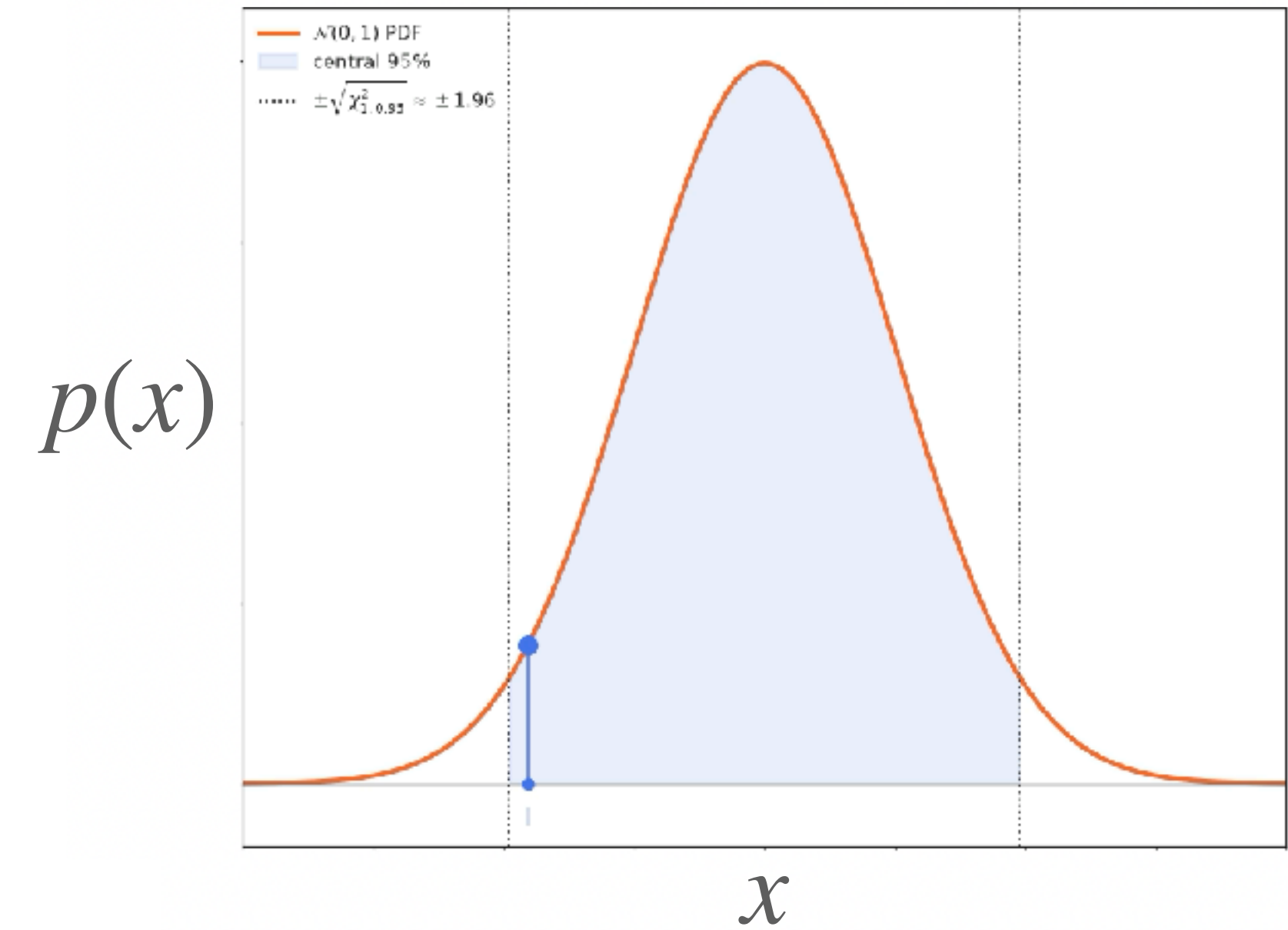


$\chi^2_{d=1}(z^2)$



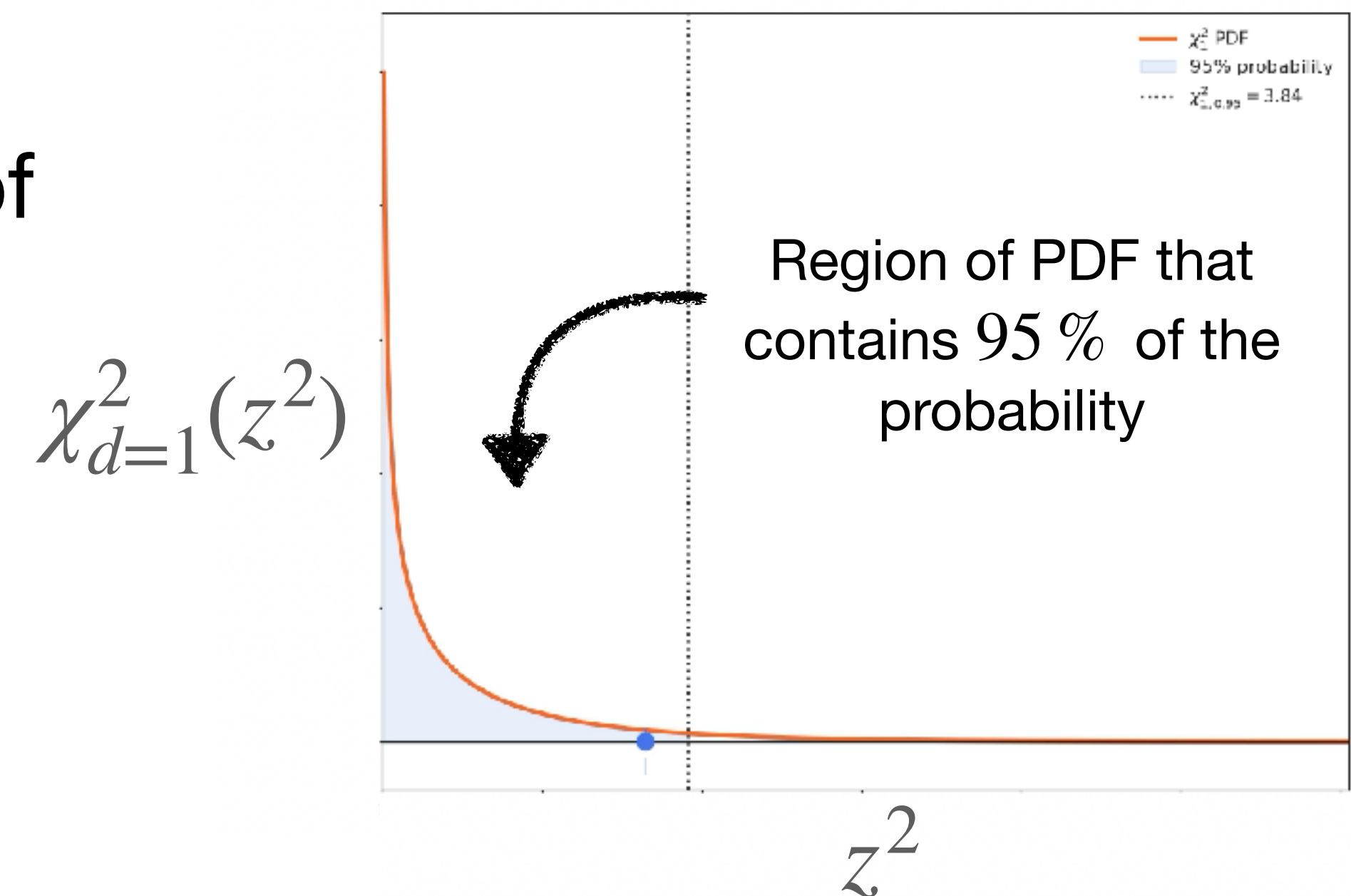
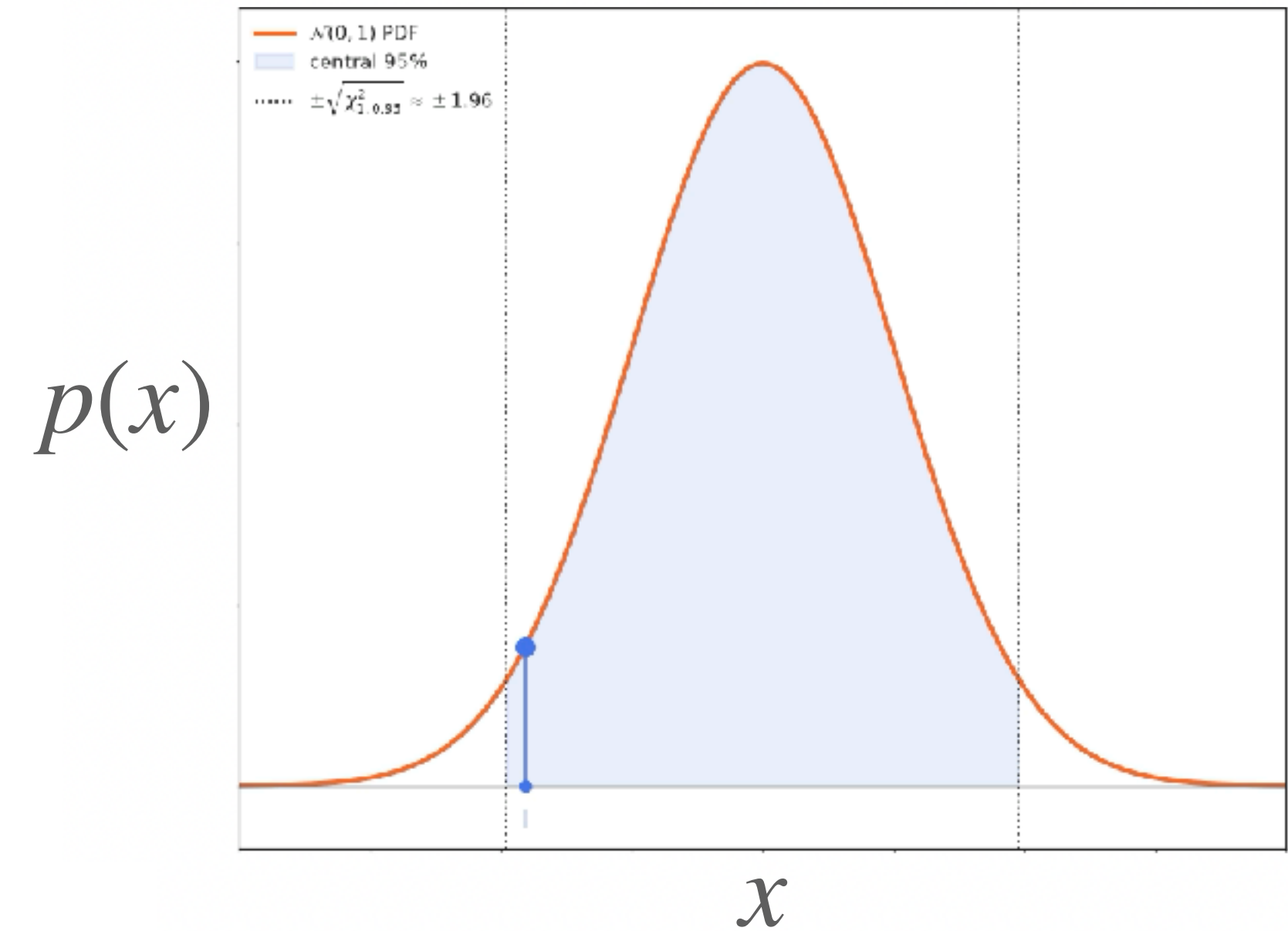
Quick detour: χ^2

- Let $x \sim \mathcal{G}[x | \mu, \sigma^2]$,
- Then $z^2 = \frac{(x - \mu)^2}{\sigma^2} \sim \chi_{d=1}^2(z^2)$.



Quick detour: χ^2

- Let $x \sim \mathcal{G}[x | \mu, \sigma^2]$,
- Then $z^2 = \frac{(x - \mu)^2}{\sigma^2} \sim \chi_{d=1}^2(z^2)$.
- The PDF $\chi_{d=1}^2(z^2)$ tells us the probability of the scatter of x around μ .



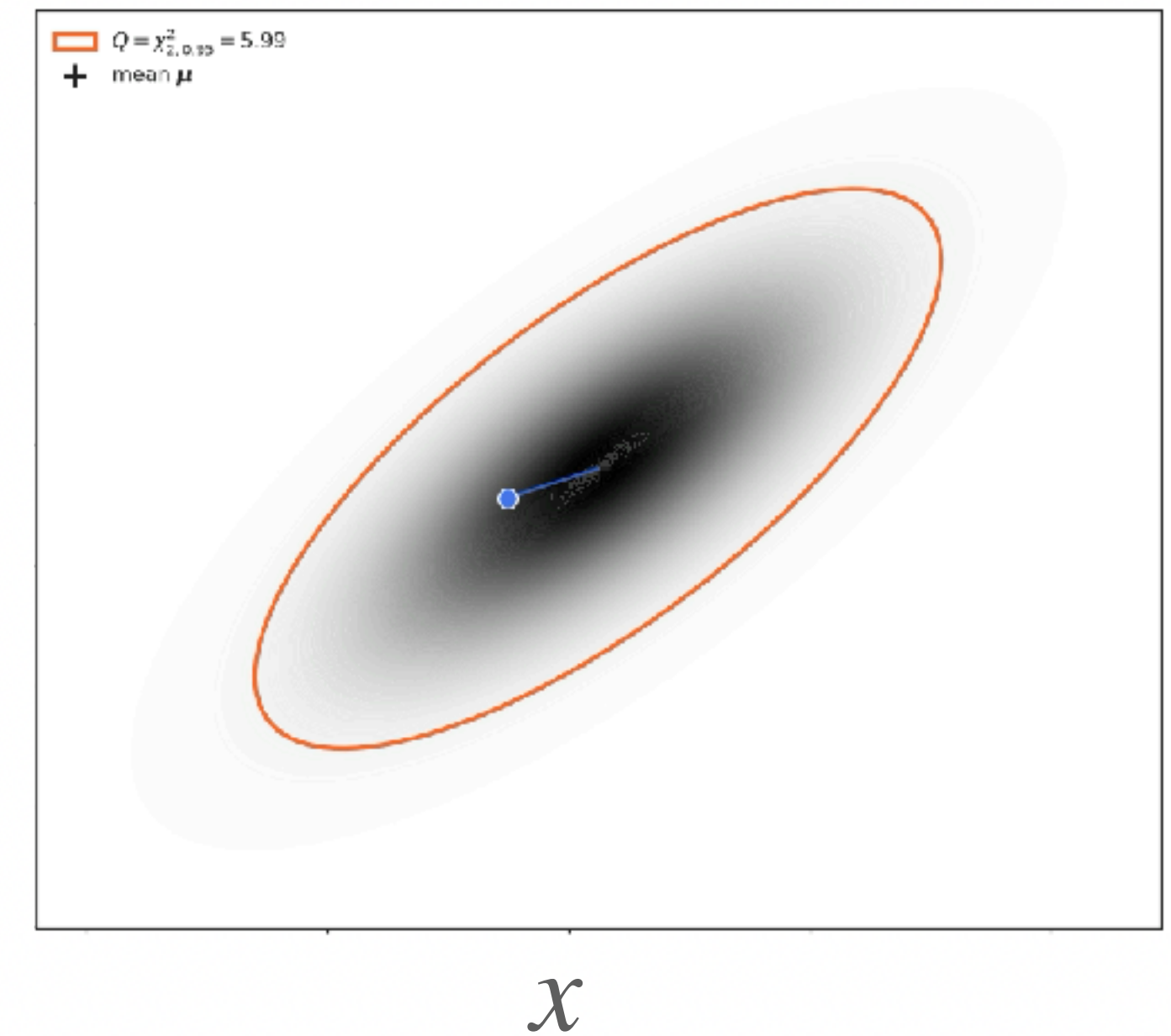
Quick detour: χ^2

- For dimension $d > 1$ where $x \sim \mathcal{G}[x | \mu, \Sigma]$ we have

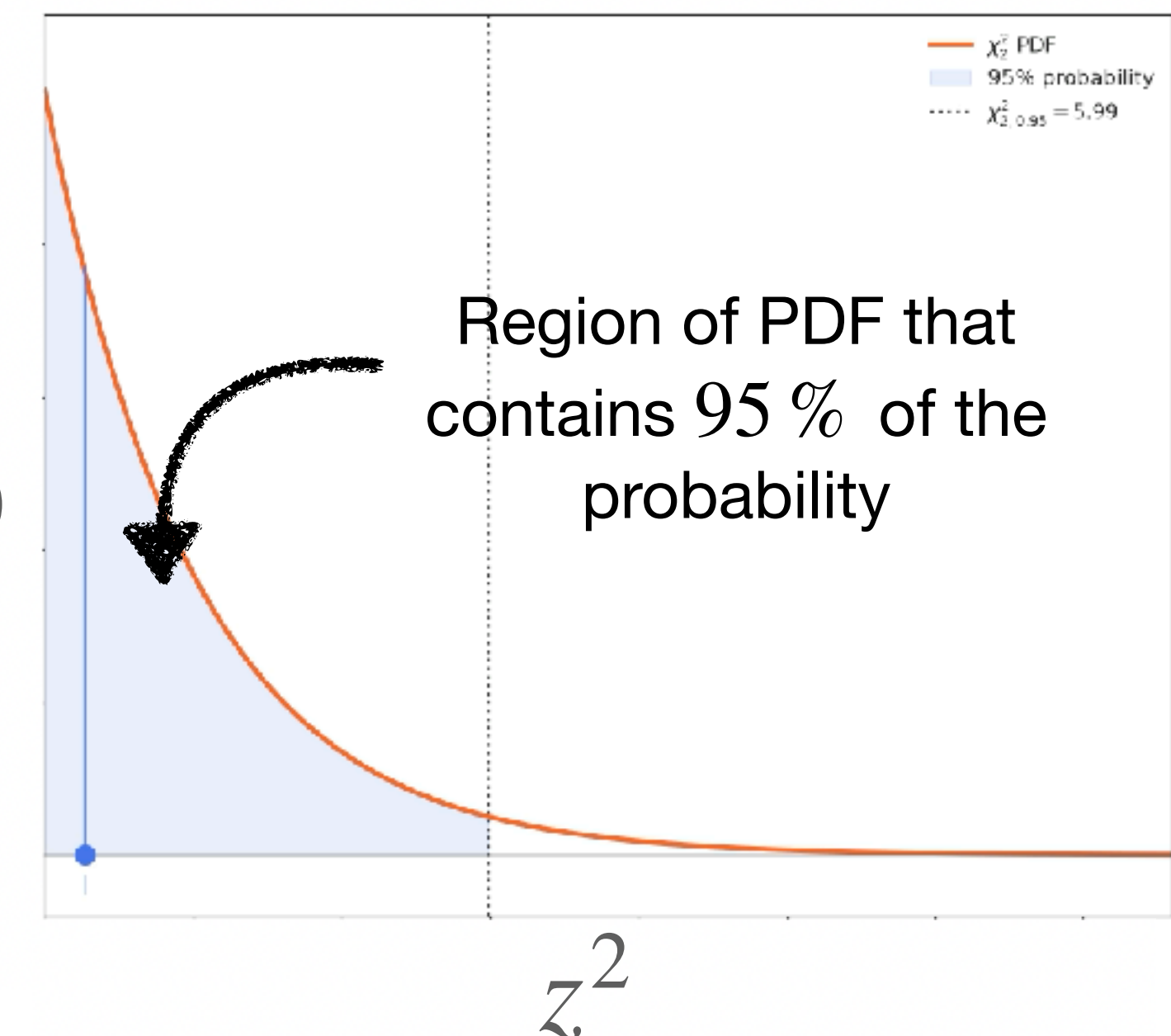
$$(x - \mu)^T \Sigma^{-1} (x - \mu) = \sum_{j=1}^d Z_j^2 \sim \chi_d^2.$$

- This follows a χ_d^2 PDF also!

$p(x)$

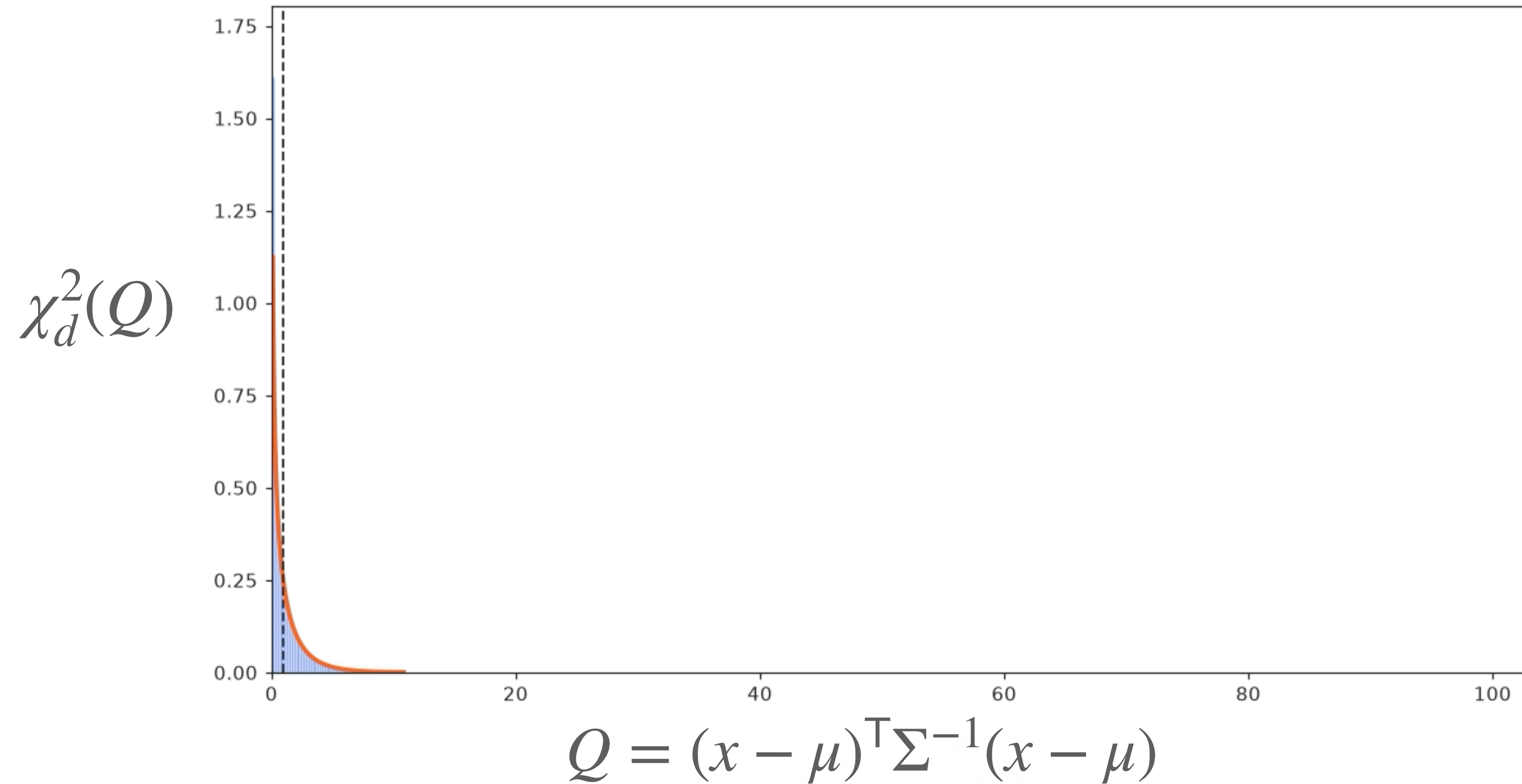


$\chi_{d=1}^2(z^2)$



Quick detour: χ^2

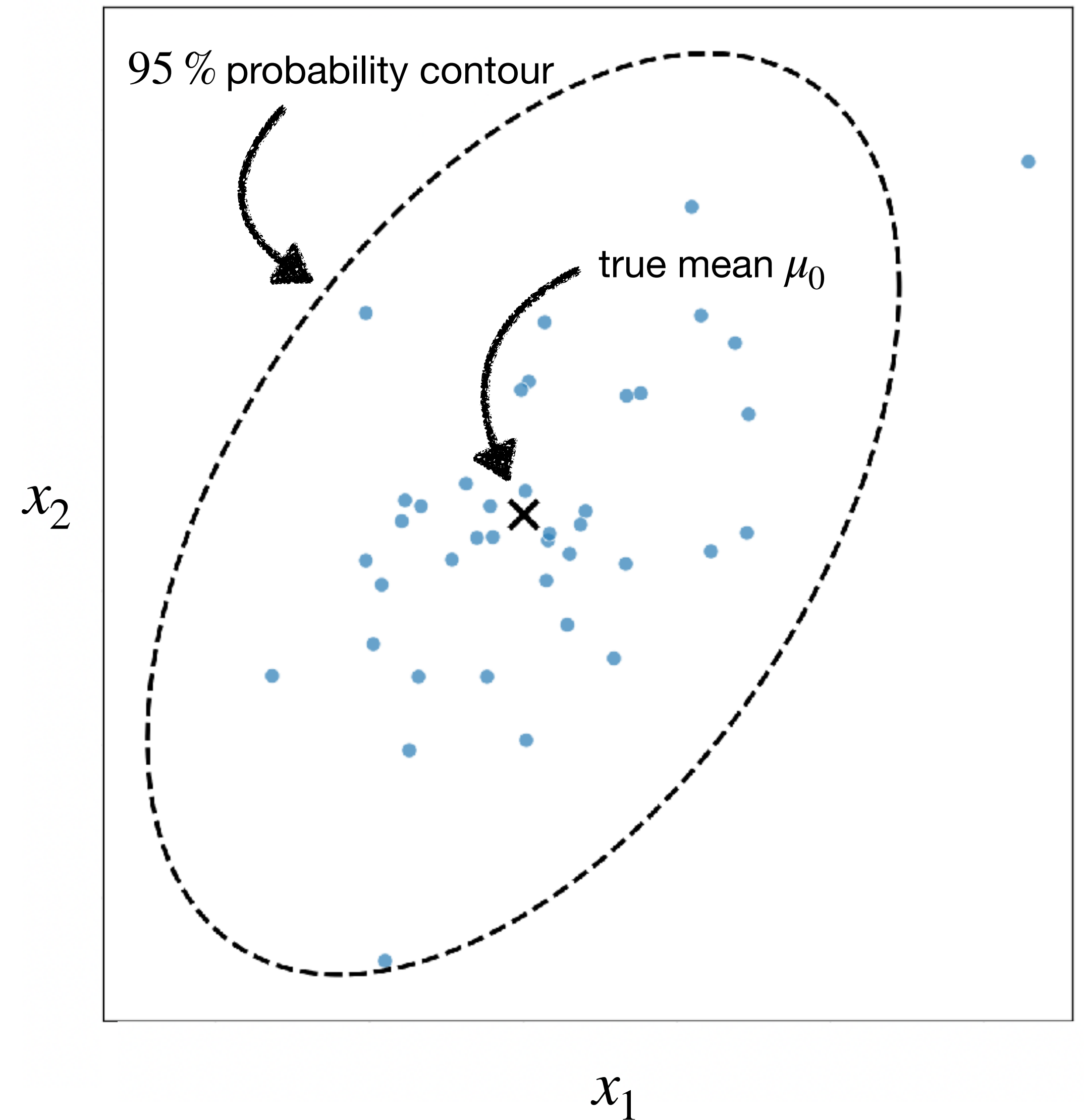
Squared Mahalanobis distances in dimension $d = 1$



Quick detour: χ^2

- So, back to the contour...

Samples $\mathcal{X} = \{x_1, \dots, x_n\} \sim \mathcal{G}(x | \mu, \Sigma)$



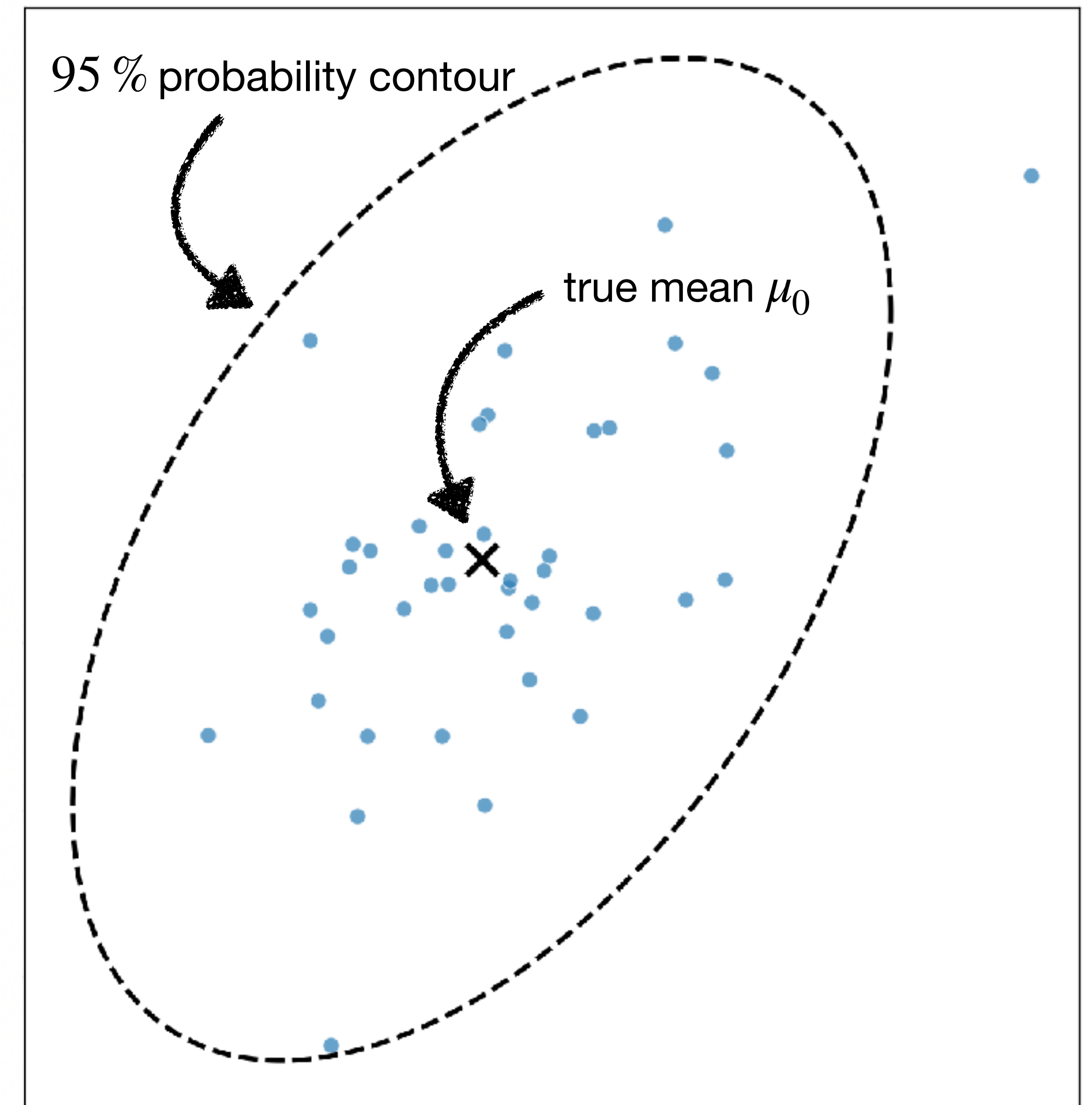
Quick detour: χ^2

- So, back to the contour...
- The scatter of x samples around μ are inside the contour in 95% of repeated experiments!

$$(x - \mu_0)^T \Sigma^{-1} (x - \mu_0) \leq \chi_{d=2,0.95}^2$$

- You may be familiar with the " 2σ region"

Samples $\mathcal{X} = \{x_1, \dots, x_n\} \sim \mathcal{G}(x | \mu, \Sigma)$



Problem

- Design an estimator $\hat{\mu}$ and calculate its sampling distribution $p(\hat{\mu} | \mu)$.
- We will use $\hat{\mu}_{\text{MLE}} = \bar{x}$,
- In this case, our estimator PDF is $p(\hat{\mu}_{\text{MLE}} | \mu) = \mathcal{G}[\hat{\mu}_{\text{MLE}} | \mu, n^{-1}\Sigma]$.

Frequentist

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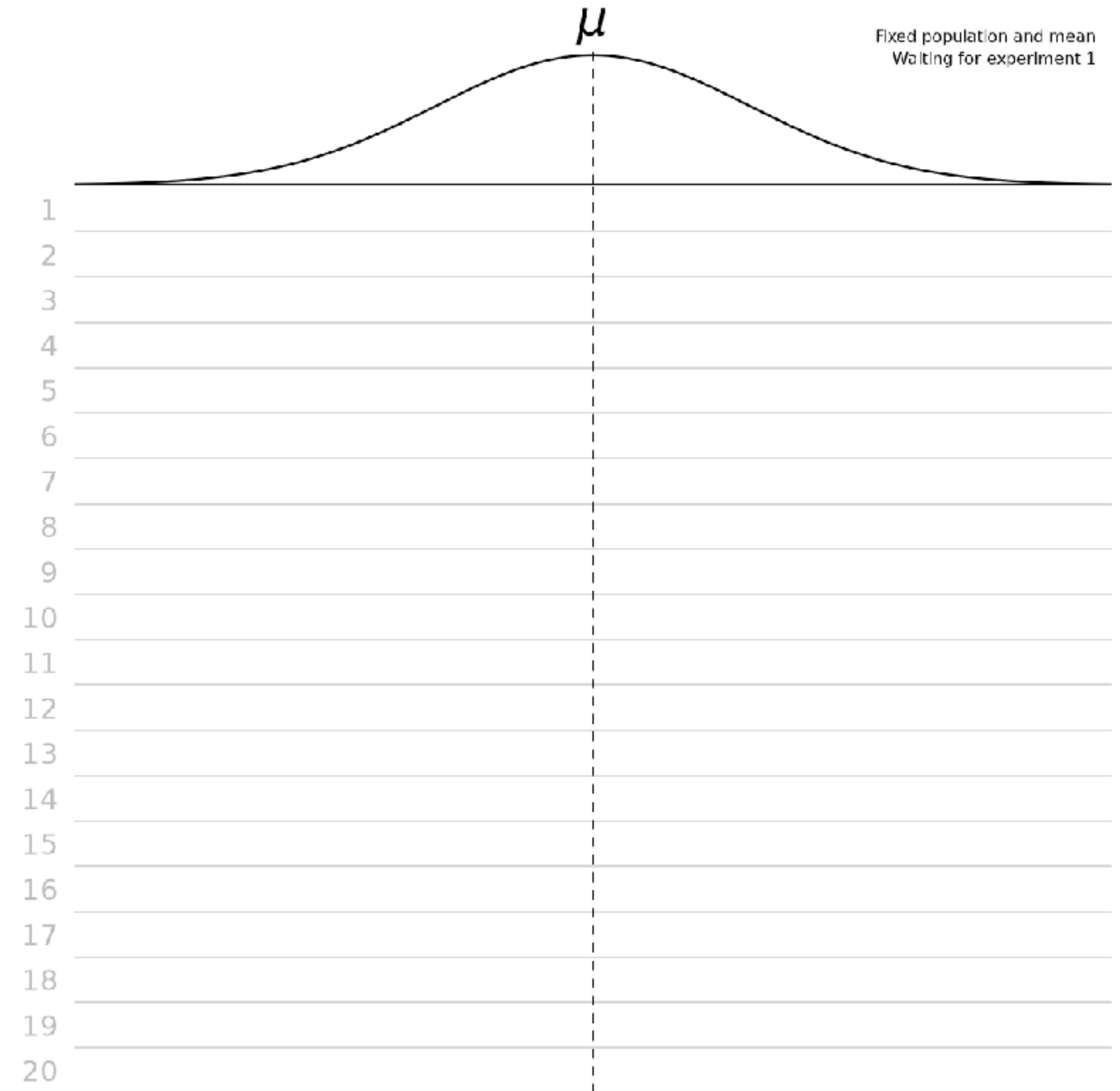
The χ_d^2 distribution

- The value $\chi_d^2(\mu)$ is the distance of μ from $\hat{\mu}$, weighted by Σ^{-1} .
 - This is random for a given experiment!

$$\begin{aligned}\Delta\chi^2(\mu) &= \chi^2(\mu) - \chi^2(\hat{\mu}) \\ &= n(\hat{\mu} - \mu)^T \Sigma^{-1} (\hat{\mu} - \mu)\end{aligned}$$

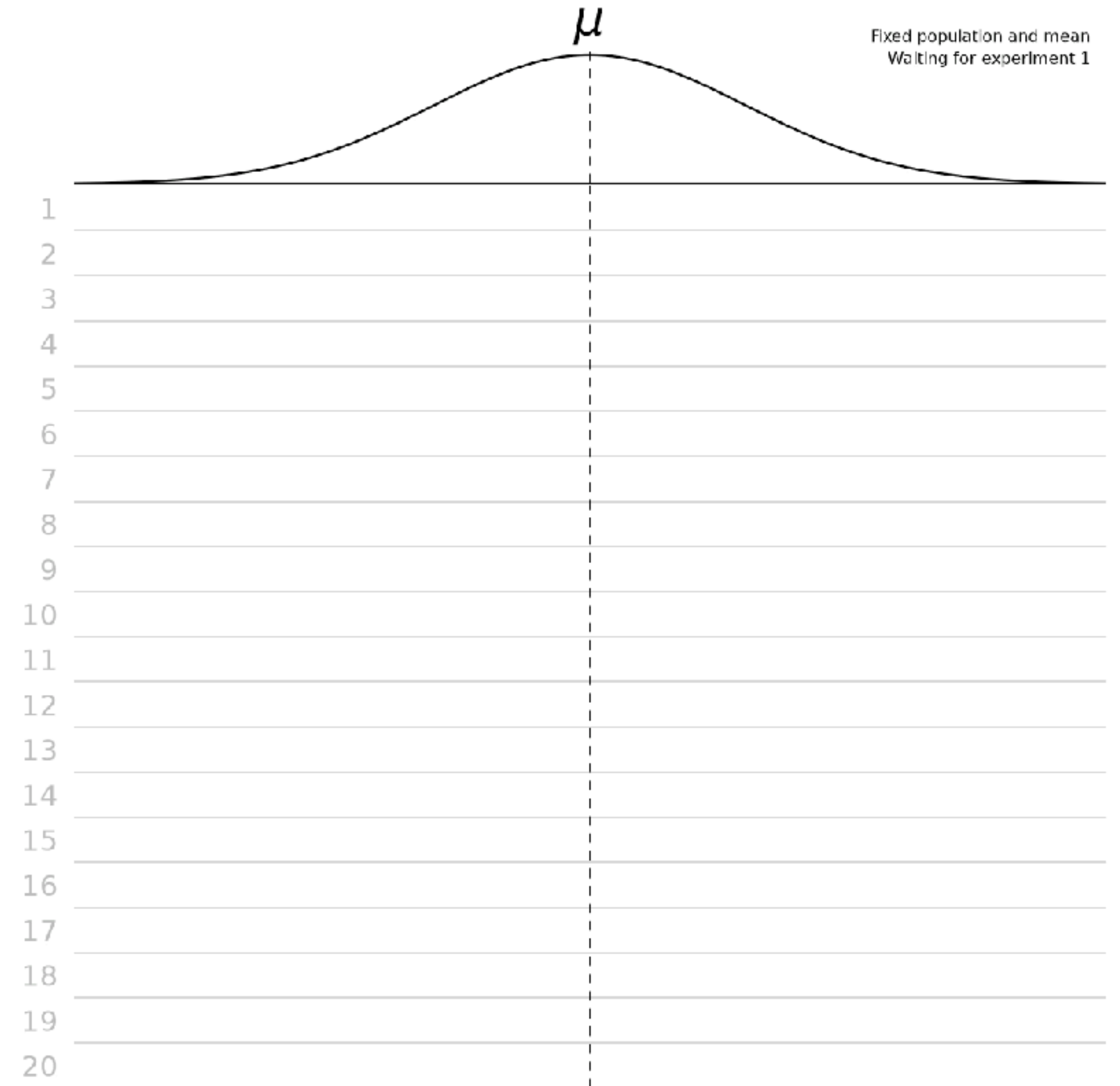
The Frequentist Confidence Interval (CI)

- What is the region of parameter space that recovers the true parameter in some fraction (e.g. 95%) of repeated experiments?



The Frequentist Confidence Interval (CI)

- What is the region of parameter space that recovers the true parameter in some fraction (e.g. 95%) of repeated experiments?
- We can define this using $\hat{\mu}$, n , Σ and $\chi^2_{d,0.95}$.
 - All these objects are constants.
 - We do *not* need to know the true μ here.



Repeated experiments for $\hat{\mu}$ when μ and σ^2 are unknown

Constructing the confidence interval

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- Calculate the $\chi^2_{d,1-\alpha}$ value (a constant given d).
- Calculate the values around $\hat{\mu}$ where $\Delta\chi^2(\mu) \leq \chi^2_{d,1-\alpha}$.
- This gives $\mathcal{C}_{0.95}^{\text{CI}} = \{ \mu : n(\mu - \hat{\mu})^\top \Sigma^{-1}(\mu - \hat{\mu}) \leq \chi^2_{d,0.95} \}$.

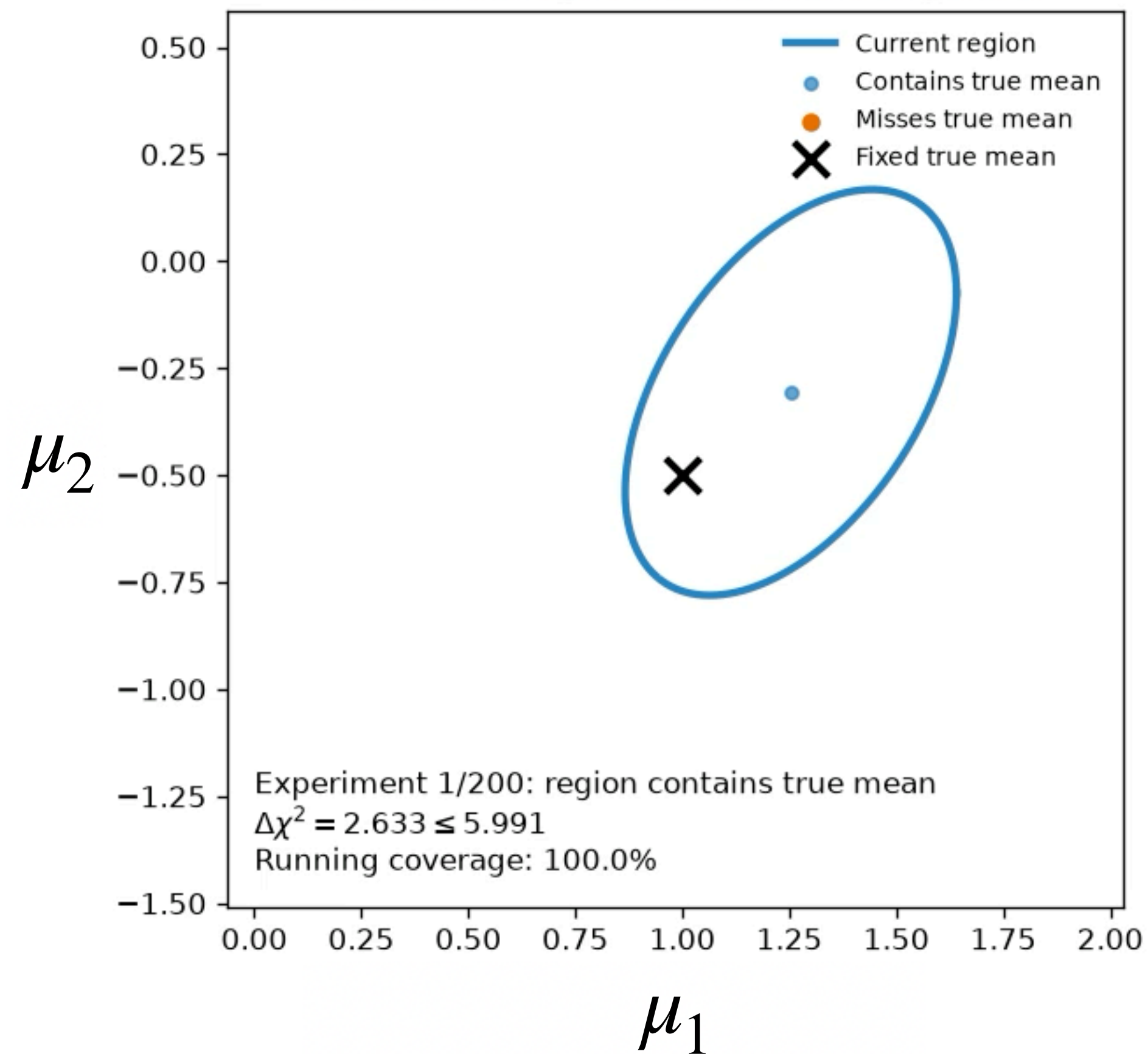
Constructing the confidence interval

- In the Frequentist interpretation of probability our result is...
- *"The interval below contains the true value μ in 95% of repeated experiments"*
-

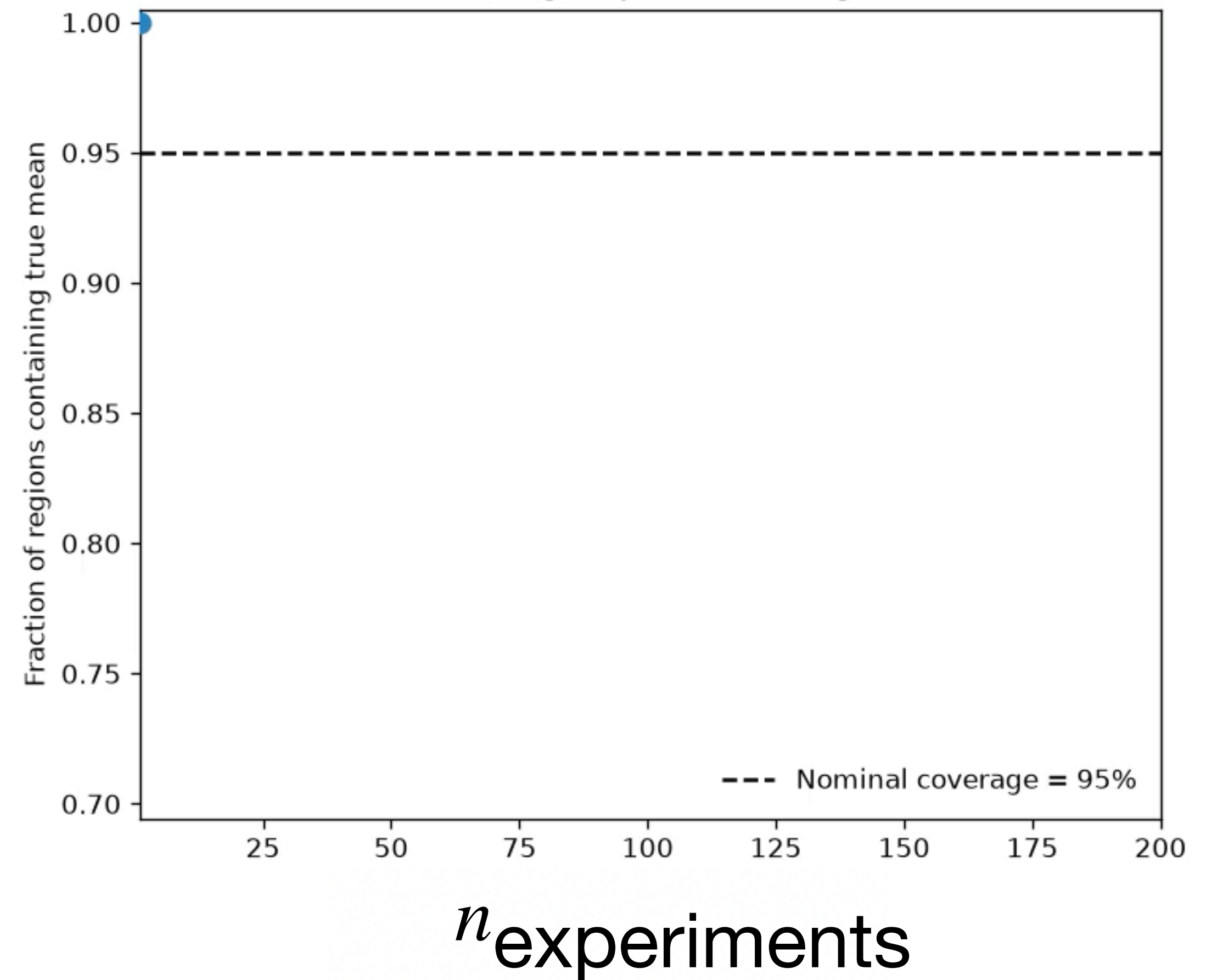
$$P \left[n \left(\hat{\mu}_{\text{MLE}} - \mu \right)^{\top} \Sigma^{-1} \left(\hat{\mu}_{\text{MLE}} - \mu \right) \leq \chi_{d,0.95}^2 \right] = 0.95.$$

Example: The confidence interval $\mathcal{C}_{0.95}^{\text{CI}}$ in repeated experiments

95% confidence regions given
independent $\hat{\mu}$ from experiments

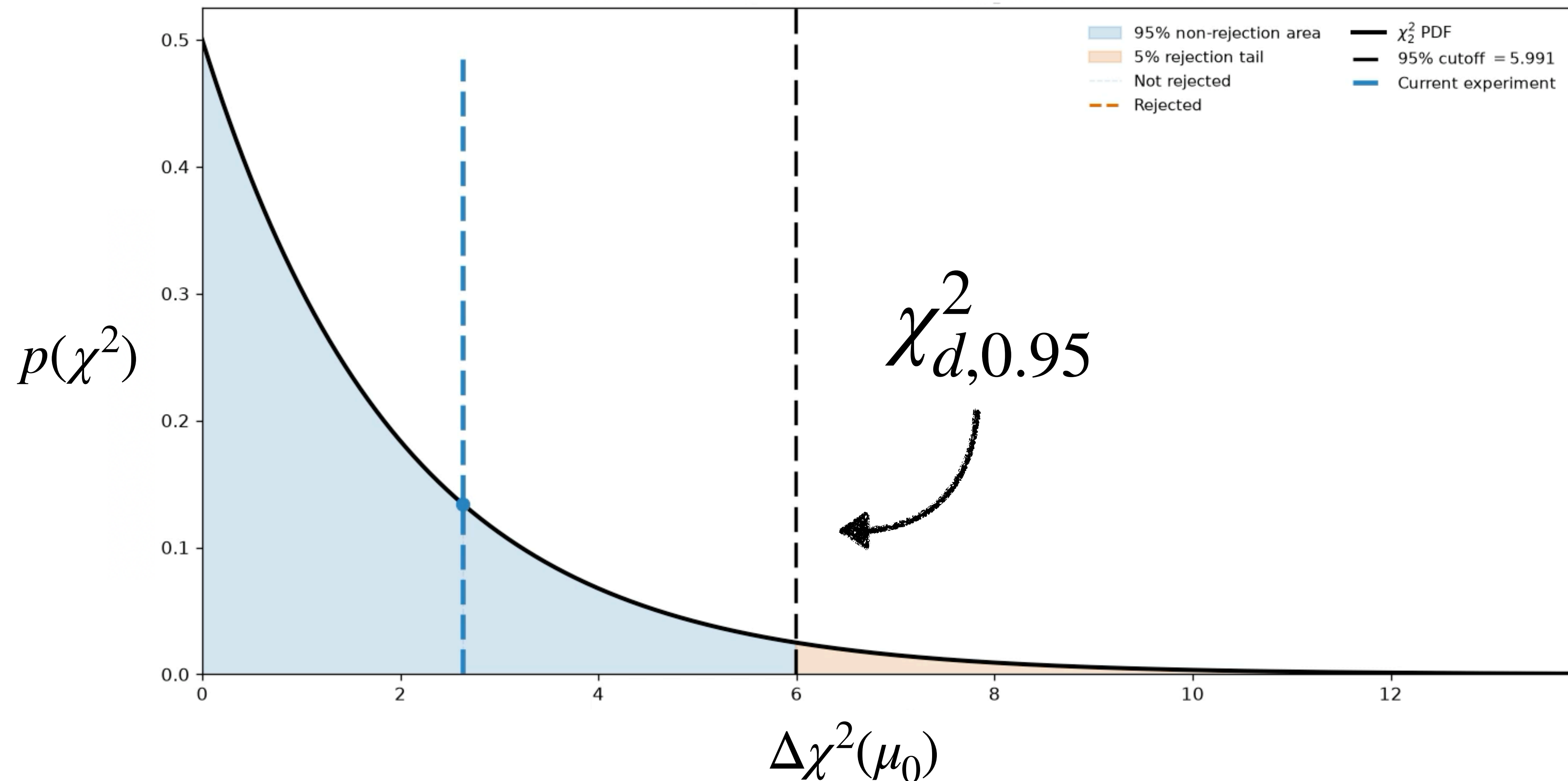


95% coverage fraction over
independent experiments



Example: The $\chi^2(\mu)$ value in repeated experiments

Experiments with independent $\Delta\chi^2(\mu) = \chi^2(\mu) - \chi^2(\hat{\mu})$



The Frequentist analysis

- The Frequentist measured a set of data $\mathcal{X}$, built a (perfectly accurate) model $\mathcal{G}[x | \mu, \Sigma]$.

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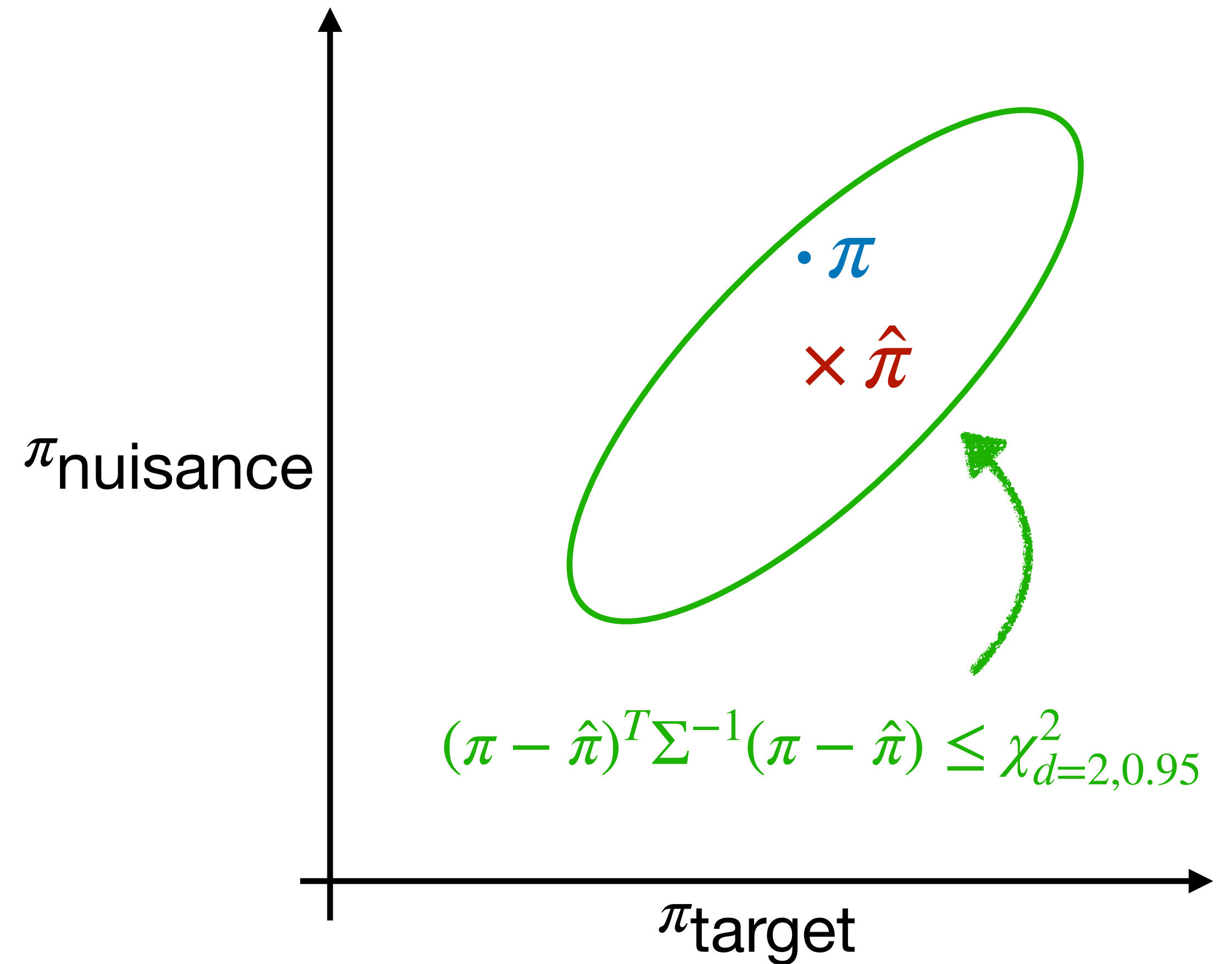
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- They defined an estimator $\hat{\mu}$ which is unbiased and has minimum variance
 - The estimator had a known distribution $p(\hat{\mu} | \mu) = \mathcal{G}[\hat{\mu} | \mu, \Sigma/n]$
- Using the estimator $\hat{\mu}$, they defined a region of μ -space within which they claim the true μ falls in 95% of repeated experiments.

Problems with the Frequentist analysis

- What if we didn't know σ^2 and we had to marginalise over it?

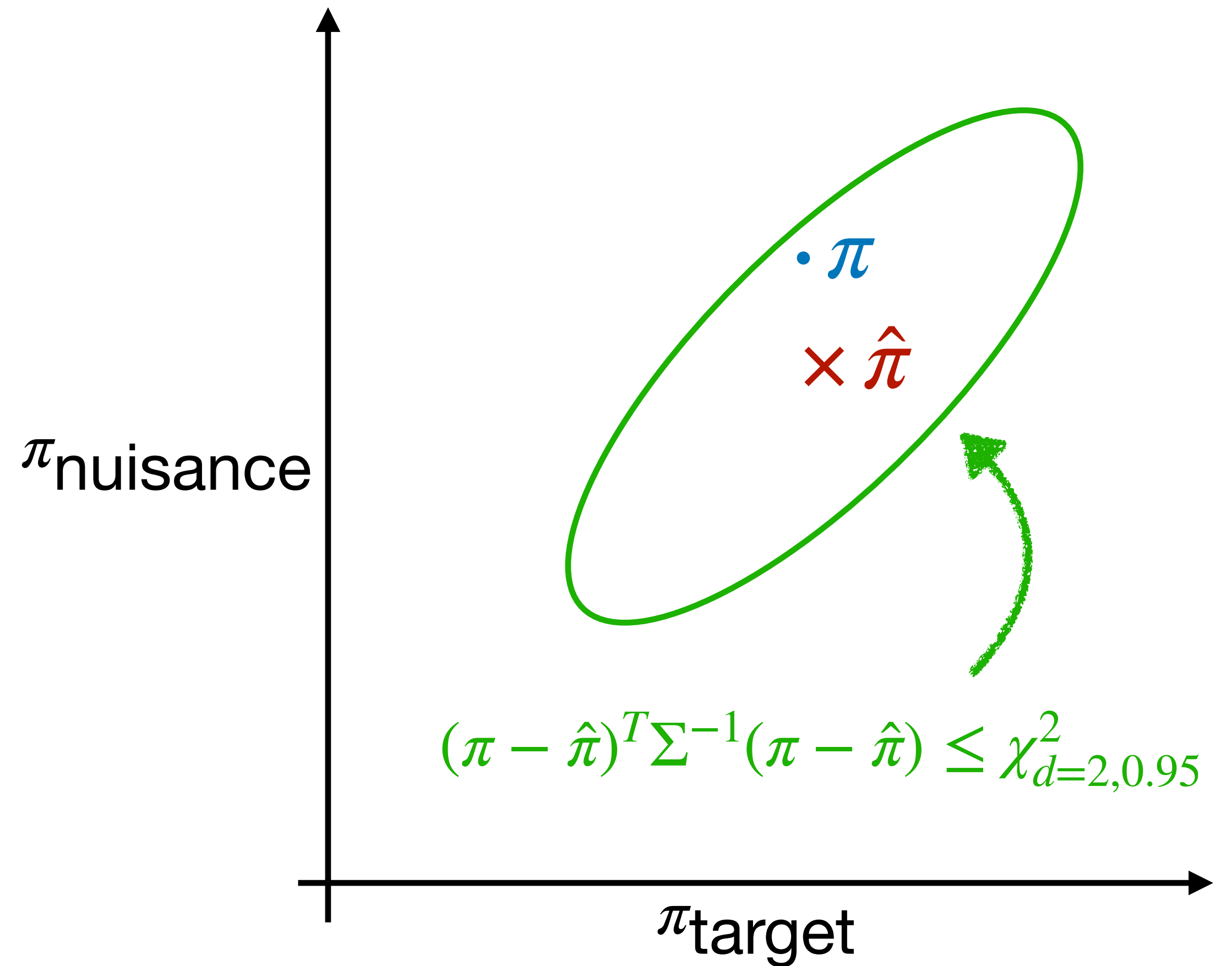
- Example of a *nuisance* parameter, e.g.

$$\pi = \{\pi_{\text{target}}, \pi_{\text{nuisance}}\}.$$



Problems with the Frequentist analysis

- What if we didn't know σ^2 and we had to marginalise over it?
- Example of a *nuisance* parameter, e.g.
 $\pi = \{\pi_{\text{target}}, \pi_{\text{nuisance}}\}$.
- To marginalise over σ^2 we treat it as a random variable!
- This is *not* what Frequentists assume!



Write down something you learned today and
something you need to hear more about next lecture

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